ILM API

Date: 04/04/2026, 17:15
Runs per endpoint: 5
Methodology: Sequential, warm cache

Performance Benchmark Report

NestJS API Migration — Escola, Municipio, Turma, Ciclo modules

Platforms Tested

Fly.io

URL: https://ilm-api-prod.fly.dev
Region: gru (Sao Paulo)
Plan: shared-cpu-1x, 512MB
Wins: 6/10 endpoints

Render

URL: https://ilm-api-prod.onrender.com
Region: Ohio (US East)
Plan: Starter ($7/mo)
Wins: 0/10 endpoints

Supabase Direct

URL: https://gghgkgmgxsqkqkabtnch.supabase.co
Region: sa-east-1 (Sao Paulo)
Type: PostgREST + RLS
Wins: 4/10 endpoints

Response Times (avg of 5 runs, in ms — lower is better)

Endpoint	Fly.io (gru)	Render (Ohio)	Supabase Direct
Turma list (20)	**1110ms p50:1098 min:1094 max:1140**	1605ms p50:1522 min:1281 max:2062	8073ms p50:8062 min:8056 max:8125
Turma by escola	1314ms p50:1054 min:1051 max:2356	1720ms p50:1385 min:1368 max:2850	**260ms p50:238 min:235 max:349**
Turma by ciclo	**1224ms p50:1094 min:1086 max:1760**	1464ms p50:1386 min:1367 max:1672	8063ms p50:8056 min:8051 max:8091
Escola list (20)	**129ms p50:79 min:79 max:326**	1504ms p50:1192 min:1184 max:2532	136ms p50:113 min:111 max:222
Escola by municipio	92ms p50:77 min:70 max:157	1185ms p50:1183 min:1180 max:1195	**68ms p50:58 min:56 max:112**
Municipio list	77ms p50:63 min:59 max:136	698ms p50:676 min:670 max:798	**63ms p50:63 min:59 max:66**
Single municipio	60ms p50:44 min:43 max:123	312ms p50:287 min:285 max:404	**57ms p50:55 min:53 max:66**
Ciclo list	**42ms p50:42 min:39 max:47**	286ms p50:286 min:282 max:291	59ms p50:57 min:55 max:68
Livros ativos	**52ms p50:52 min:50 max:54**	543ms p50:543 min:540 max:548	57ms p50:56 min:55 max:60

Endpoint	Fly.io (gru)	Render (Ohio)	Supabase Direct
Health check	**53ms** p50:41 min:40 max:104	285ms p50:286 min:281 max:288	88ms p50:82 min:80 max:100

Data Parity

Entity	API Count	Supabase Count	Match
turmas	1229	N/A	Diff (RLS)
escolas	281	281	Exact
municipios	38	38	Exact
ciclos	4	4	Exact

Key Findings

- **Fly.io (gru)** is the fastest for complex queries (turma list with relations, turma by ciclo) due to same-region DB access (~5ms DB round-trip)
- **Supabase Direct** is fastest for simple queries from this test machine (direct connection, no API overhead) — but from Brazilian users, Fly.io gru would be faster
- **Render (Ohio)** is consistently slowest due to cross-region latency to Supabase sa-east-1 (~450ms per DB round-trip)
- The NestJS API eliminates 80+ RLS policy evaluations per query, which is why turma queries are dramatically faster through the API
- Data parity is exact for escolas (281), municipios (38), and ciclos (4)

Generated by ILM API Benchmark Suite. Test executed from Windows machine (external to all platforms). For Brazilian end-users, Fly.io gru latency would be ~20-50ms lower than shown (same country, shorter network path). Supabase direct numbers represent browser-to-PostgREST latency and include RLS policy evaluation time.